

## Subject Description Form

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| <b>Subject Code</b>                                   | COMP5523                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Subject Title</b>                                  | Computer Vision and Image Processing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Credit Value</b>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Level</b>                                          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Pre-requisite/<br/>Co-requisite/<br/>Exclusion</b> | Nil                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Objectives</b>                                     | <ul style="list-style-type: none"> <li>• To let the students learn the fundamental principles on the aspects of interdisciplinary research including acquiring, processing, analyzing, understanding and utilizing high-dimensional visual data from the real world;</li> <li>• To equip the students with the knowledge of how to develop artificial intelligent systems which automate tasks that the human visual system can do;</li> <li>• To guide the students to understand the relevant state of art technologies and gain experience throughout a variety of case studies.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Intended Learning Outcomes</b>                     | <p>Upon completion of the subject, students will be able to:</p> <ol style="list-style-type: none"> <li>a. critically review and consolidate various methods for digital image processing and analysis and relate or apply them to different applications.</li> <li>b. grasp various algorithms for vision related tasks and develop creative solutions to practical problems by considering various requirements and complex issues (e.g., legal/social/ethics issues).</li> <li>c. carry out in-depth analysis of the digital image data with different image data models, pattern recognition algorithms and learning theory.</li> <li>d. demonstrate leadership and qualities of reflective practitioners by engaging in a group project on vision related topics and various learning activities.</li> <li>e. communicate effectively to peers and senior colleagues and present quality deliverables through reading and writing reports and documents.</li> </ol>                                                                                                                                                                                                                                                 |
| <b>Subject Synopsis/<br/>Indicative Syllabus</b>      | <ol style="list-style-type: none"> <li>1. Introduction               <ol style="list-style-type: none"> <li>a. The background</li> <li>b. An overview of the related fields.</li> <li>c. The development of vision systems</li> </ol> </li> <li>2. Image acquisition and formation models               <ol style="list-style-type: none"> <li>a. Imaging systems</li> <li>b. Geometry of image formation</li> <li>c. Camera models and calibration</li> <li>d. Orthographic and perspective projections</li> </ol> </li> <li>3. The fundamentals of image processing               <ol style="list-style-type: none"> <li>a. Image enhancement</li> <li>b. Image representation</li> <li>c. Image transformations</li> </ol> </li> <li>4. The fundamentals of deep learning               <ol style="list-style-type: none"> <li>a. Basic concept of machine learning</li> <li>b. Deep neural network</li> <li>c. Transformer</li> </ol> </li> <li>5. 2D Vision               <ol style="list-style-type: none"> <li>a. Image feature, matching and retrieval</li> <li>b. Image segmentation, understanding and object recognition</li> <li>c. Optical flow, video tracking and motion detection</li> </ol> </li> </ol> |

|                                                                                | <div><div>6. d. Image and video generation<br/>3D vision</div><div>a. The geometry of multiple views</div><div>b. Reconstruction: Shape from multiple views and single view.</div><div>c. 3D Scene understanding</div><div>7. Applications: Industry, mapping, robots and games.</div><div>8. Professionalism and legal/social/ethics issues related to computer vision (e.g., personal data (privacy) ordinance)</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                |   |   |   |                                                                                                 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
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| Teaching/<br>Learning<br>Methodology                                           | <div>Lectures provide students with the content of main concepts and methods of the course, together with comprehensive examples, and class questions/answers/discussions for easy understanding.</div> <div>Tutorials and lab sessions offer the opportunity for students to review and consolidate the lecture and reference materials through exercises and software tools.</div> <div>Project assignments will give students the opportunity to solve practical visual information analysis problems.</div> <div>Written assignments help students to develop a solid foundation of visual information processing and analysis throughout the project work of case studies.</div>                                                                                                                                                                                                                                                                                                                                         |                                                                                |   |   |   |                                                                                                 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| Assessment<br>Methods in<br>Alignment with<br>Intended<br>Learning<br>Outcomes | <table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="5">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th></tr><tr><td>1. Written assignments and quiz</td><td>15</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Project Assignment</td><td>15</td><td></td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>3. Examination</td><td>70</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100</td><td colspan="5"></td></tr></table> <div>Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:</div> <div>Written assignments, tests/quizzes, and examination are used to test and deepen students’ understanding of the subject materials while project assignment is used to provide a hands-on experience of how the subject materials are applied.</div> |                                                                                |   |   |   |                                                                                                 | Specific assessment methods/tasks | % weighting | Intended subject learning outcomes to be assessed (Please tick as appropriate) |  |  |  |  | a | b | c | d | e | 1. Written assignments and quiz | 15 | ✓ | ✓ | ✓ | ✓ | ✓ | 2. Project Assignment | 15 |  | ✓ | ✓ | ✓ | ✓ | 3. Examination | 70 | ✓ | ✓ | ✓ |  |  | Total | 100 |  |  |  |  |  |
| Specific assessment methods/tasks                                              | % weighting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Intended subject learning outcomes to be assessed (Please tick as appropriate) |   |   |   |                                                                                                 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | a                                                                              | b | c | d | e                                                                                               |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| 1. Written assignments and quiz                                                | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓                                                                              | ✓ | ✓ | ✓ | ✓                                                                                               |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| 2. Project Assignment                                                          | 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                | ✓ | ✓ | ✓ | ✓                                                                                               |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| 3. Examination                                                                 | 70                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ✓                                                                              | ✓ | ✓ |   |                                                                                                 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| Total                                                                          | 100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                |   |   |   |                                                                                                 |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |
| Student Study<br>Effort Expected                                               | <div>Class contact:</div> <div><div>▪ Lectures</div><div>▪ Tutorials/Labs</div></div> <div>Other student study effort:</div> <div><div>▪ Reading</div><div>▪ Written and Project Assignments, Quizzes/Tests,</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                |   |   |   | <div></div> <div>26Hrs.</div> <div>13Hrs.</div> <div></div> <div>26Hrs.</div> <div>40Hrs.</div> |                                   |             |                                                                                |  |  |  |  |   |   |   |   |   |                                 |    |   |   |   |   |   |                       |    |  |   |   |   |   |                |    |   |   |   |  |  |       |     |  |  |  |  |  |

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|                                    | Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |
|                                    | Total student study effort                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>105Hrs.</b> |
| <b>Reading List and References</b> | <p>1. D.A. Forsyth and J. Ponce, <i>Computer Vision: A Modern Approach</i>, 2<sup>nd</sup> Ed., Prentice-Hall, 2011.</p> <p>2. Gonzalez, Rafael, <i>Digital Image Processing</i>, 4<sup>th</sup> Ed., Pearson Hall, 2019, ISBN-13: 978-9353062989 ISBN-10: 9353062985</p> <p>3. Richard Szeliski, <i>Computer Vision: Algorithms and Applications</i>, 2<sup>nd</sup> Ed., 2022, Springer Publisher, ISBN-13: 978-3030343712 ISBN-10: 3030343715.</p> <p>4. Goodfellow, Ian, et al. Deep learning. Vol. 1. No. 2. Cambridge: MIT press, 2016.</p> <p>5. Rafael C. Gonzalez. Digital Image Processing, 4Th Edition. 2018.</p> <p>6. V Kishore Ayyadevara and Yeshwanth Reddy. Modern Computer Vision with PyTorch: A practical roadmap from deep learning fundamentals to advanced applications and Generative AI, 2<sup>nd</sup> Ed., 2024, ISBN-13: 978-1803231334 ISBN-10: 1803231335.</p> <p>7. Proceedings of Proceedings of IEEE Conference on Computer Vision and Pattern Recognition (CVPR).</p> <p>8. IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI).</p> <p>9. Relevant articles from other conferences and/or journals such as IJCV, ICCV, and ECCV.</p> |                |